

Comparison of Eye-Tracking Representations for Engagement Inference in Office Workers

Tomi Božak
Jožef Stefan Institute
Ljubljana, Slovenia
tb85088@student.uni-lj.si

Gašper Slapničar
Jožef Stefan Institute
Ljubljana, Slovenia
gasper.slapnicar@ijs.si

Abstract

Eye tracking is an appealing sensor for inferring momentary user states at work, but before any model is trained a practitioner must decide how to represent a short gaze window. We compare four representation families – raw multichannel signals, handcrafted ocular features, frozen GazeMAE embeddings and frozen MOMENT embeddings – on identical three-second windows from an ongoing office-worker study, with self-reported momentary engagement as the target. All four are evaluated under leave-one-subject-out cross-validation with fold-local preprocessing with random forest classifier and an MLP and are compared against a majority baseline. Raw signals give the best unseen-participant performance (0.538 accuracy, 0.014 above the baseline), handcrafted features follow closely, and neither frozen encoder improves on them. Absolute discrimination is weak throughout (best ROC-AUC 0.551) and between-participant variability dwarfs the differences between representations, so we report the ranking as preliminary, cost-aware guidance rather than a settled result.

Keywords

eye tracking, engagement inference, representation learning, GazeMAE, MOMENT, subject generalisation

1 Introduction

Interactive systems that adapt to the person using them need evidence about that person’s momentary state. In knowledge work, engagement—how absorbed someone is in the task at hand—is an attractive target: it fluctuates within a working day, it is only weakly visible in application logs, and it is the state most directly tied to productivity and well-being interventions. Eye tracking is a plausible sensor for it. Gaze position, fixation and saccade dynamics, blinks and pupil size are available continuously from a screen-mounted tracker, require no action from the user, and have documented links to attention and processing load.

Turning a stream of gaze samples into model input, however, forces a design decision that is rarely examined on its own. A short window of eye-tracking data can be presented to a classifier in at least four ways. The raw multichannel signal retains everything the tracker measured, at the price of high dimensionality and no structure. Handcrafted ocular features—fixation, saccade, blink and pupil statistics—encode decades of domain knowledge, stay interpretable, and remain the default choice in applied work [3, 2]. Learned representations promise to remove the manual step: GazeMAE [1] provides gaze-specific embeddings pretrained

on eye-movement corpora, and general-purpose time-series foundation models such as MOMENT [4] provide a frozen encoder intended to transfer to unseen domains. The four options differ substantially in preprocessing effort, storage and compute, and practitioners have little evidence on which to base the choice.

That evidence gap exists because the families are seldom compared like for like. Published results differ in windows, labels, classifiers and, most consequentially, in validation protocol, so an apparent advantage of one representation can just as easily come from an optimistic split. What an applied designer needs is the comparison under the condition that matters for deployment: a model trained on some people and applied to a person it has never seen.

We therefore ask which eye-tracking representation supports the most reliable inference of self-reported momentary engagement in unseen office workers. We answer with a controlled comparison of raw signals, handcrafted features, GazeMAE embeddings and MOMENT embeddings on identical three-second windows, an identical binary engagement target and identical leave-one-subject-out folds. Both pretrained encoders are kept frozen and only a prediction head is trained, which is the out-of-the-box setting a practitioner would try first. Our contributions are (i) a leakage-safe, subject-disjoint comparison of the four representation families on spontaneous office work, and (ii) cost-aware guidance for representation selection under limited labelled data.

2 Related Work

Eye tracking has long been used as behavioural evidence for internal user states. Pupil dilation is a classical correlate of processing load [5], and fixation, saccade, blink and pupil statistics computed over short windows remain the dominant input to models of workload, affect and engagement. Our own prior work illustrates both the appeal and the limits of this pipeline: handcrafted ocular features supported coarse affective distinctions [3], yet a systematic evaluation across several prediction targets showed that task type is recognised far more reliably than graded workload, and that reported accuracy depends strongly on how the prediction target and the validation split are defined [2]. Feature engineering encodes useful domain knowledge, but the resulting performance is bounded by what a fixed feature set captures.

Learned representations offer an alternative. GazeMAE pretrains temporal autoencoders on raw position and velocity signals and shows that the resulting gaze-specific embeddings support downstream classification through a simple linear head [1]. In parallel, general-purpose time-series foundation models such as MOMENT provide a frozen multichannel encoder intended to transfer across domains without task-specific pretraining [4]. Both are presented as replacements for hand-designed features, and both are typically evaluated on curated benchmarks rather

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

Information Society 2026, Ljubljana, Slovenia

© 2026 Copyright held by the owner/author(s).

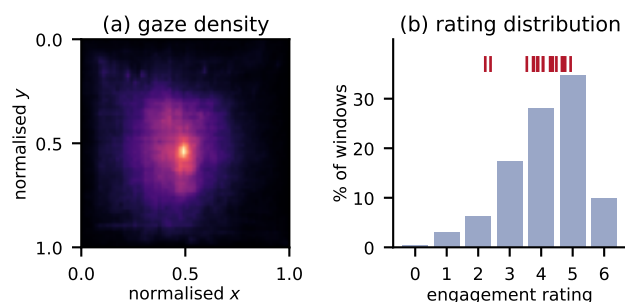


Figure 1: Descriptive context for the modelling cohort. (a) Gaze-coordinate density on the normalised display, pooled over all on-screen samples. (b) Distribution of the engagement rating over the 140,531 modelled windows; the red ticks mark the 17 individual participant means, which span 2.2–4.9 and motivate the within-participant centring of the target.

than on modest, subject-dependent recordings of spontaneous behaviour. How much of that benefit survives when such encoders are applied out of the box in this setting remains open.

Evaluation design is a further concern. Windows drawn from the same person are highly dependent, so random window splits and preprocessing fitted on the full dataset inflate reported performance, an effect documented in detail for other biosignal domains [6]. Comparisons across representations are interpretable only when every candidate is evaluated on identical windows and labels under a subject-disjoint protocol with fold-local preprocessing.

We therefore extend our earlier feature-based analyses to learned representations. This preliminary study is an empirical comparison rather than a new encoder or fine-tuning method: raw signals, handcrafted features, GazeMAE and MOMENT embeddings are evaluated on the same windows, the same engagement target and the same leave-one-subject-out folds.

3 Methodology

3.1 Data and Eye-Tracking Representations

The data come from an ongoing multimodal study in which office workers are recorded during ordinary work at their own workplace. The wider collection comprises an RGB camera, a screen-mounted eye tracker, a microphone and screen-activity logs; this paper uses the eye-tracking modality only. The dataset is neither complete nor public, so this is a preliminary study on the cohort available at the time of writing: 17 participants and 657 full-workday recordings, between 19 and 79 recordings per participant (median 32). The tracker samples at 60 Hz, and we retain five channels: left and right pupil size, gaze x and y , and the average eye–screen distance.

Labels come from an experience-sampling protocol. Participants are prompted several times per working day with a short questionnaire covering engagement, affect, fatigue, stress, physical comfort, workplace interactions, sleep and productivity. Our target is the absorption item, “Right now I am immersed in my work”, one of three momentary engagement items answered at every prompt and rated on a seven-point scale from 0 to 6; we call it the *engagement rating*. Each prompt labels the 15 minutes of recording that precede it, and only those intervals enter this study.

Recordings are segmented into three-second windows with a 2.25-second step (25 % overlap). Every window carries an identifier built from the participant, the source recording and the position of the window inside that recording; the four representation exports are joined on this identifier, so all models see exactly the same windows. Windows with too little usable gaze are discarded before embedding (at least 30 % valid frames and at least 32 valid frames), and remaining gaps inside a retained window are linearly interpolated. Of roughly 7.3 million recorded windows, only those inside a labelled interval are usable; 177,244 of them yield handcrafted features, 157,685 also pass the encoder quality control, and 140,531 carry a numeric engagement rating. That final cohort—about 2 % of all recorded windows, and between 2,136 and 25,447 windows per participant—is what we model. Using labelled windows only discards a great deal of recorded data, but it keeps a preliminary experiment tractable and the target unambiguous.

Display resolutions range from 1920×1080 to 3840×2160 pixels. They differ between participants and, for a few participants, between that person’s own recordings, so gaze coordinates are normalised to $[0, 1]$ using the resolution declared in each source recording rather than one value per participant. Out-of-screen coordinates are retained for modelling; 94.7 % of samples fall on the display, and only those enter the descriptive density map in Figure 1(a), which shows the expected central concentration together with broad coverage of the screen.

Four representations are computed from these windows. **Raw** keeps the five channels, truncates or pads each to 120 samples and flattens them into a 600-dimensional vector. **Handcrafted features** start from 199 exported columns; 18 identifier, timestamp and questionnaire columns are removed structurally so that no survey metadata can leak into the model, leaving 181 numeric candidates covering blink counts and durations, fixation and saccade statistics, pupil size and velocity moments, eye–screen distance statistics and per-channel missingness. **GazeMAE embeddings** come from the public micro–macro autoencoder checkpoints applied to gaze position and velocity resampled to 500 Hz, with the 128-dimensional position and velocity codes concatenated into 256 dimensions [1]. **MOMENT embeddings** come from the frozen MOMENT-1-LARGE encoder applied to the same five channels with an input length of 512 and mean pooling, giving 1,024 dimensions [4]. Both encoders are used out of the box; no weights are updated.

3.2 Experimental Design

Engagement ratings are strongly person-specific: the participant means span 2.2 to 4.9 on the 0–6 scale (Figure 1(b)). An absolute threshold would therefore mostly encode who the participant is rather than when they were engaged, so we binarise relative to each person’s own level. Inside every fold, the ratings of each training participant are mean-centred separately, and the held-out participant’s ratings are centred with the unweighted mean of the training participants’ means; a window is positive when its centred rating exceeds zero. Over the cohort this gives 56.3 % positive windows, ranging from 33 % to 72 % per participant.

The primary protocol is leave-one-subject-out cross-validation over the 17 participants. Everything data-dependent is fitted inside the fold: handcrafted-feature filtering (dropping columns with more than 60 % missing values, zero variance, or absolute Pearson correlation above 0.8 with a retained column, which leaves 58–60 of the 181 candidates), median imputation, and

Table 1: Leave-one-subject-out results for binary engagement inference, averaged over the 17 held-out participants. SD is the standard deviation across participants and Δ is the improvement in accuracy over the majority baseline, paired within folds. Balanced accuracy is relative to 0.500 by construction. Accuracy uses all 17 folds; the remaining metrics use the 16 folds whose test participant is not single-class. M-F1 is macro-F1. Best value per column in bold.

	Acc.	SD	Δ	Bal. acc.	M-F1	AUC
Majority baseline	.524	.346	—	.500	.326	.500
<i>Raw signals (600-d)</i>						
Random forest	.527	.147	+ .003	.529	.460	.546
MLP	.538	.113	+ .014	.536	.468	.551
<i>Handcrafted features (58–60-d)</i>						
Random forest	.528	.230	+ .005	.508	.418	.527
MLP	.531	.145	+ .008	.521	.451	.535
<i>GazeMAE embeddings (256-d)</i>						
Random forest	.535	.155	+ .012	.509	.441	.513
MLP	.520	.107	− .004	.493	.435	.494
<i>MOMENT embeddings (1,024-d)</i>						
Random forest	.521	.286	− .002	.501	.380	.504
MLP	.507	.274	− .017	.497	.362	.504

standardisation are all estimated on the training participants and then applied unchanged to the held-out participant.

Three predictors are run on every representation: a most-frequent baseline, a random forest (300 trees, balanced class weights) and an MLP (1028–512–256–128–64 units, GELU activations, layer normalisation, Adam with learning rate 2×10^{-4} and weight decay 10^{-3} , batch size 1024, at most 200 epochs with early stopping after 15 epochs without improvement). A single seed (42) is used throughout. The representation $\times$ model grid is a robustness check on the representation ranking rather than a model comparison.

We report accuracy as the primary metric, together with its improvement over the baseline paired within folds, and balanced accuracy, macro-F1 and ROC-AUC as secondary discrimination measures; all are aggregated as unweighted means over participants. One participant rated every window below the training centre, so that test fold contains a single class: accuracy is defined for all 17 folds and the remaining metrics for 16. As a contextual check we also run a one-step persistence predictor within recordings, carrying the last observed label forward in rolling blocks of 300 calibration and 90 test windows; it is a different, within-participant protocol and is not comparable with the main comparison.

4 Results

Table 1 reports the main comparison. Every representation–model combination sits close to the majority baseline. The best result is the raw signal with the MLP, at 0.538 accuracy, 0.014 above the paired baseline, and it is also best on balanced accuracy (0.536), macro-F1 (0.468) and ROC-AUC (0.551). Handcrafted features come next and improve on the baseline with both classifiers (up to 0.531 accuracy and 0.535 AUC). The two frozen encoders trail: GazeMAE reaches 0.535 accuracy with the random forest but falls to 0.520, below the baseline, with the MLP, while MOMENT is at or below the baseline on every metric and its AUC (0.504) is indistinguishable from chance.

The ordering raw $\geq$ handcrafted $>$ GazeMAE $>$ MOMENT holds across all four metrics, but the absolute differences are

small—0.003 to 0.017 in accuracy—and far smaller than the spread across held-out participants, whose accuracy standard deviations range from 0.11 to 0.29. We therefore read the ordering as a descriptive tendency in this cohort and attach no significance claim to it. GazeMAE in particular warrants a flag rather than a verdict: it is the only representation whose sign changes with the classifier, whereas raw signals and handcrafted features improve on the baseline with both, so its usefulness here is an open question rather than a settled negative result. A related and tentative observation is that the MLP helps exactly those representations that carry some signal and hurts the two embedding sets; one possible reading is that a more flexible head pays off only when there is something to fit, but the effect is small and we do not build on it.

Figure 2 gives a geometric view of the same difficulty. In two-dimensional t-SNE projections none of the four representations arranges windows by engagement rating; the colours are mixed everywhere, in the learned embeddings as much as in the raw signal. Colouring the identical projections by participant instead reveals some local grouping for raw signals and GazeMAE, suggesting that what structure these representations do capture is at least partly person-specific rather than state-specific.

For context, a one-step persistence predictor applied within recordings reaches 0.999 accuracy, 0.960 balanced accuracy and 0.965 macro-F1 over 1,434 temporal blocks. This is a sanity check on the label series rather than a competitive result: because each questionnaire answer is broadcast unchanged to every window of the preceding 15-minute interval (Section 3), most consecutive window pairs share a label by construction. It is not comparable with the unseen-participant results above.

5 Discussion

To answer the research question directly: on this cohort the raw multichannel window is the most informative of the four representations for binary engagement inference in unseen office workers, with handcrafted features close behind and the two frozen embeddings improving on neither. The margin over the majority baseline is, however, small for every candidate, so the more robust reading of Table 1 is that three seconds of gaze does not make momentary self-reported engagement well predictable across people, whichever representation is used. The representation effect, modest as it is, is still larger and more consistent than the classifier effect, which reorders nothing except the sign of the two embedding results.

The failure of the frozen encoders to improve on simpler inputs is best read as a mismatch between pretraining and task. GazeMAE was pretrained on controlled stimulus-viewing corpora and MOMENT on generic time-series benchmarks; neither has seen day-long, spontaneous office gaze with its dropouts, idle stretches and heterogeneous displays. Using them out of the box is therefore not a verdict on representation learning for eye tracking, but it is the first thing a practitioner tries, and in this setting it does not pay off. Domain adaptation or fine-tuning on in-domain gaze, rather than frozen inference, is the natural next step—particularly for GazeMAE, whose classifier-dependent behaviour (Section 4) leaves its usefulness here genuinely open.

For practice, raw signals are the strict empirical winner and also the cheapest option: no feature engineering, no pretrained checkpoint, no GPU inference pass, and 600 dimensions against MOMENT’s 1,024. We would not generalise this into a recommendation to abandon handcrafted features. They finished a close

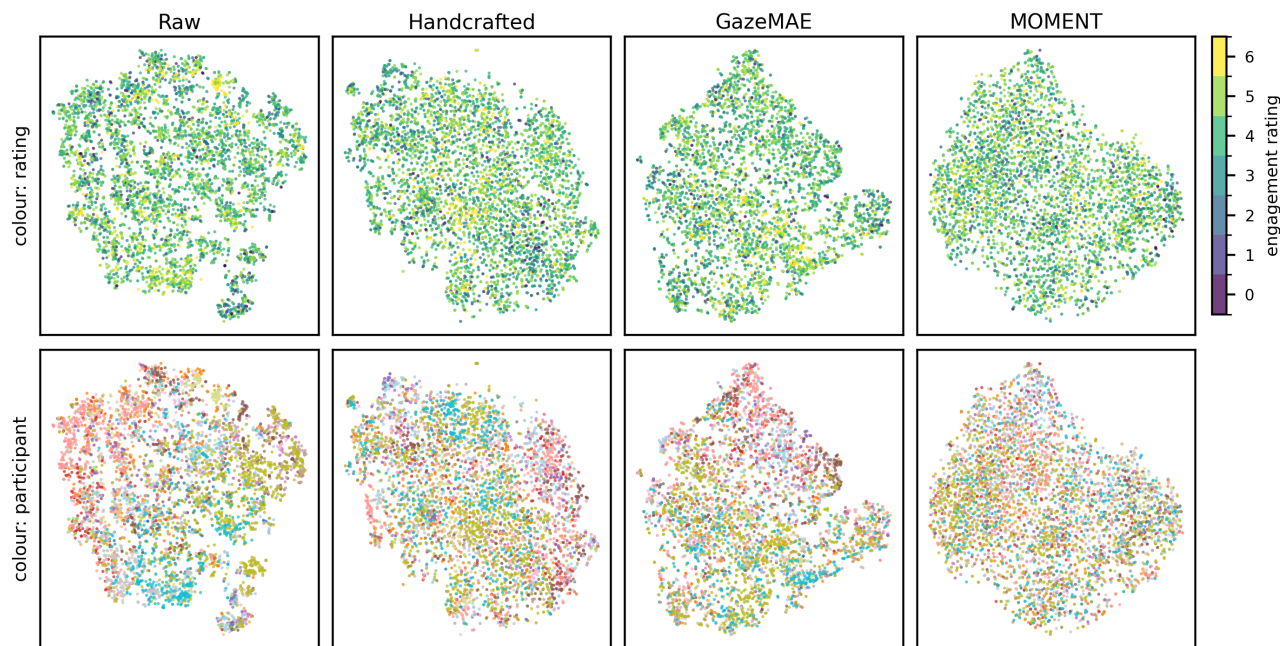


Figure 2: t-SNE projections of the four representations over a common sample of 5,000 windows, coloured by engagement rating (top) and by participant (bottom). No representation organises the windows by rating; the mild local grouping visible for raw signals and GazeMAE follows participants rather than states. The projections are exploratory geometry and are not evidence about separability under a trained classifier.

second, they remain interpretable, and our earlier feature-based analyses found them effective on related eye-tracking targets [3, 2], so they are still worth trying on a similar problem. What we would not recommend is reaching for a frozen general-purpose encoder as a default.

Several limitations bound these conclusions. The study uses one unpublished and still-growing dataset with 17 participants; the target is a noisy self-report assigned retrospectively to a 15-minute interval and binarised by within-participant centring; and the results depend on one windowing scheme, one seed, one set of quality-control rules and two specific public checkpoints. The density map and the projections are descriptive only. Future work should validate the comparison on further datasets, adapt or fine-tune the encoders on in-domain gaze, explore calibrated personalisation for new users, and test longer windows and aggregated targets that match the granularity at which engagement is actually reported.

6 Conclusion

We compared four eye-tracking representations—raw multichannel windows, handcrafted ocular features, frozen GazeMAE embeddings and frozen MOMENT embeddings—under one identical, leakage-safe leave-one-subject-out protocol for momentary engagement inference in office workers. The simplest and cheapest representation was also the strongest, reaching 0.538 accuracy and 0.551 ROC-AUC on unseen participants, and neither pre-trained encoder repaid its additional cost out of the box. For practitioners working with limited labelled data, the evidence here favours starting from raw windows or handcrafted features and treating frozen foundation-model embeddings as something to be adapted before they are trusted.

Acknowledgements

[To COMPLETE: funding, grant numbers and acknowledgements]

References

- [1] Louise Gillian C. Bautista and Prospero C. Naval Jr. 2021. GazeMAE: general representations of eye movements using a micro-macro autoencoder. In *2020 25th International Conference on Pattern Recognition (ICPR)*, 7004–7011. DOI: 10.1109/ICPR48806.2021.9412761.
- [2] Tomi Božak, Shivalika Goyal, Marc Langheinrich, Martin Gjoreski, and Gašper Slapničar. 2026. Evaluating feature-based machine-learning models with post hoc explainability for eye-tracking-based task type and workload inference. *AI*, 7, 8, 325. DOI: 10.3390/ai7080325.
- [3] Tomi Božak, Mitja Luštrek, and Gašper Slapničar. 2024. Feature-based emotion classification using eye-tracking data. In *Proceedings of the 27th International Multiconference Information Society (IS 2024), Volume A: Slovenian Conference on Artificial Intelligence*. Ljubljana, Slovenia. <https://is.ijs.si/?p=16837>.
- [4] Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. 2024. MOMENT: a family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. <https://arxiv.org/abs/2402.03885>.
- [5] Daniel Kahneman and Jackson Beatty. 1966. Pupil diameter and load on memory. *Science*, 154, 3756, 1583–1585. DOI: 10.1126/science.154.3756.1583.
- [6] Matthew Rosenblatt, Link Tejavibulya, Rongtao Jiang, Stephanie Noble, and Dustin Scheinost. 2024. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications*, 15, 1, 1829. DOI: 10.1038/s41467-024-46150-w.